

this sort of information, there is another tool that has become a very useful part of my toolkit now, which is called *iftop*. It's a pretty old tool, and it hasn't been updated in a couple of years, but don't let that stop you from trying it.

iftop is an interactive program with a number of cool features, all of them accessible by typing some key, and it has a quick 1-screen online help in case you forget the keys. Running *iftop -nNPB* on a lightly-loaded system might look like the output shown in Figure 5.

The display is quite self-explanatory, except for the last three columns in the main display. These are averages of the data transferred over the previous 2, 10 and 40 seconds respectively.

The black bars are important. Across the very top is the 'scale' for all the bars, and the bars actually represent the 10-second average (the middle column) by default, although pressing "B" will cycle between 2, 10, and 40-second averages. This way you get a visual indication of what hosts and ports are hogging the traffic.

You can do some cool things here—you can choose to look only at outgoing or incoming traffic or perhaps the sum of the two (press *t* to cycle between these modes). You can aggregate all traffic for each source into one line by pressing *s*, and for each destination by pressing *d*. Be sure to read the online help as well as the man page -- it's worth it.

What's even more cool is that there are two filters to limit the output. Typing *l* enables a 'display filter'—the pattern or string you enter will be applied to the host+port field and used to filter lines appearing in the display. This is a literal match: for example, if you type "pop3" as the search filter, then use "N" to disable port number resolution, you'll have to change the search string to ":!10" in order for it to match. The same goes for host names versus IP addresses.

Using *l* only affects the display; the totals still count all the traffic. On the other hand, you can use *f* to set a packet filter condition that will stop traffic that does not match, from even coming into the program. For instance, you can type "f" then "port 25" to see only SMTP traffic. This filter can take quite complex conditions, using the same syntax that popular tools

like *tcpdump*, etc. use. Plus, this filter can be specified from the command line too, like:

```
iftop -nBP -f 'port 22'
```

All in all, this is a pretty nice tool to keep an eye on things once in a while or perhaps when someone complains things are a little slow.

iptraf is also a very nice and easy-to-use tool, with a very neat curses GUI. It actually has a lot more features than *iptraf*: the IP interface monitor shows you TCP and UDP separately, and for TCP it shows you packet flags (making it easy to identify connection attempts that are not succeeding, for instance). Overall statistics for each interface are also available in a separate screen, and on the whole it's almost a real GUI (using curses), with menus and sub-menus, etc. It also has a very slick filter specification GUI, if you're not the command line type.

Despite all this, however, I find myself using *iftop* for day to day use, because *iptraf* lacks the aggregation, multiple averages, quick and easy filtering, etc. that *iftop* does. Plus, most of the filters I want are much easier to type into *iftop* or at the command line.

Some last words

I don't use these tools every day, but when I needed them, they were really useful. Could I have gotten by without knowing them? Maybe... but that's not how we think, is it? A workman needs as many tools as he can get his hands on, and these are some of mine.

Among these, *iftop* remains the one I use more often than the others. It's actually closer to monitoring than troubleshooting, but they're all great tools, and exploring them gives you an understanding of what's happening under the hood as your machine goes about its daily business. 

By: Sitaram Chamarty

The author has a 'packrat' mentality when it comes to finding and learning about all sorts of tools, both well known and obscure. He 'carries' these tools around with him on a small USB stick, thanking God every day that he is not, for instance, a carpenter :-). Sitaram works for TCS, in Hyderabad, and can be reached at sitaramc@gmail.com

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